Saryarqa Go · Cost and Payback Sheet

ZERTTEX Hackathon 2026 · Case 1 Technology Product · all amounts in KZT · planning assumptions, not a forecast · same model and numbers as Cost_Payback_Sheet.xlsx (live formulas) and the prototype's Impact page

1. One-time costs: build + validation (Phase 1) Proposal

Item	Months	KZT
Senior full-stack developer (contract)	M1–3	2,100,000
UX/UI design finishing (freelance)	M1–2	400,000
Localization ZH + EN, native proofreading	M2–3	600,000
Legal: offer, refund rules, privacy, data consent	M1–3	450,000
Payment provider onboarding + real test payments	M2–3	150,000
Guide videos + tour photos (4 guides, 4 tours)	M2–3	500,000
Pre-launch security review	M3	250,000
Interviews + usability tests	M1–3	250,000
One-time total		**4,700,000**

2. Launch, monthly operations, reserve (year 1)

Item	Months	KZT
Xiaohongshu content + creator collaborations	M4–12	1,200,000
Printed postcards + QR cards	M4	100,000
Hosting, CDN, database, domain, e-mail, monitoring	M1–12	300,000
Chinese-speaking support, part-time	M4–12	1,080,000
Contingency reserve (assumed fully spent)	M3–12	620,000
Launch + operations + reserve		**3,300,000**
Phase 1 total, 12 months Proposal		**8,000,000**
Case cap Case fact · Phase 2, only after the month-9 stage gate		20,000,000 · ≤ 12,000,000
Per month from month 13: hosting & tools 25,000 + chinese-speaking support, part-time 120,000 + developer retainer for fixes 100,000 + marketing 150,000 Assumption		395,000

Not paid from the budget (stated): unpaid team work; Saryarqa Routes staff (product owner, manager, guides); per-transaction payment fees (inside the margin). Reasons for every line: Cost_Payback_Sheet.xlsx → Budget_8M, column G.

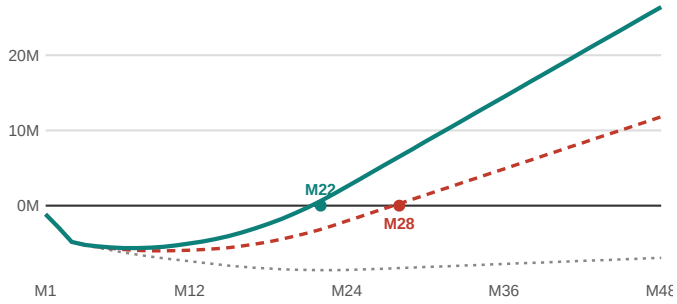
3. Revenue assumptions

Average price per traveler	44,000	Case fact
Contribution margin (after guides, transport, hotels, fees)	30%	Assumption
Contribution per traveler (price × margin)	13,200	Calculated
Launch month / ramp reaches 100% in month	4 / 24	Assumption
Extra foreign travelers per year at 100% (from month 24)	1,263	Calculated
Travelers / month to cover month-13 costs (395,000 ÷ 13,200)	≈ 30	Calculated

4. Payback: company target vs required worse case (−30%)

	Target 100%	−30%	Stress*
Lowest cumulative cash	−5.6M (M8)	−6.0M (M10)	−8.6M (M22)
Cumulative cash, month 24	+2.5M	−2.0M	−8.6M
Payback month (cumulative ≥ 0)	**Month 22**	**Month 28**	**none in 48**
Cumulative cash, month 48	+26.4M	+11.8M	−6.9M
Break-even travelers (8,000,000 ÷ 13,200)			607

*Stress test: 40% of the target path, 25% margin, launch 2 months late → the month-9 stage gate stops paid ads and keeps Phase 2 closed.



company target · worse case (−30%) · stress test · cumulative cash, 48 months

Margin	20%	25%	30%	35%	40%
Payback, target	M29	M25	M22	M20	M18
Payback, −30%	M47	M34	M28	M25	M23

5. Where 1,263 comes from

~4% of 3,400 travelers are foreign today Case fact	≈ 136
Keep the domestic travelers	3,264
30% foreign ⇒ total = 3,264 ÷ 0.7	≈ 4,663
30% of 4,663 = foreign travelers needed	≈ 1,399
Extra foreign travelers per year (1,399 − 136)	**≈ 1,263**

Check: 150,000,000 ÷ 3,400 = 44,118 KZT, slightly above the stated 44,000 KZT average price. The model uses 44,000 (the careful choice).

Method: each line is spent evenly over its months; travelers in month m = scenario % × 1,263 ÷ 12 × ramp(m), ramp = (m − 3) ÷ 21 for months 4–24, then 1; net = travelers × 13,200 − costs. Payback = first month the cumulative total is ≥ 0. Labels: Case fact = given in the case; Assumption / Proposal = team estimate to confirm with real quotes and company cost data.